Foreman notebook, §23

The law of a ray read off its relations: the box of the walls, the masters, the one-step matrix, and the law without closings

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This entry follows §22 of the file of §§15–22 (22–24 October 2026) and updates its Ledger: items O1 *and* O10 *of L.2, three new items* O21, O22 *and* O23*, and L.5. Numbers* 15.x–22.x *are those of that file; handbook citations are by printed number and page of the handbook PDF of 20 October 2026. Every object is a member of the serif family or a construction of the handbook, written with its slots; classical names appear only as aliases, in the last section.*

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23 The law of a ray read off its relations

The reading of a ray (Procedure 18.6) and its examination (§19) rest on a law of the ray that holds at every strength, with a certificate. Step (3) of the procedure is the one step that guesses. It computes closings, finds their least recurrence by linear algebra, reflects it into a candidate (Corollary 18.2), and then searches an ansatz for a certificate (Proposition 17.15; §19, Step 4). Neither half is a construction: sixteen closings do not prove their own recurrence (§19, Step 3), and an ansatz is a choice of denominators and supports that either contains a certificate or does not. On the three-mark centre both halves are cheap. On the eight-mark list V the first half was never completed (Proposition 20.37(vi)): a closing of V costs j^7, and a law of V's size needs hundreds of them.

The burden of this entry is to produce, from the list and the direction alone, a law that is proved as an identity in the strength, with its certificate, and to say what that costs. The ingredients are already in the handbook. The values of all insertions along a ray form one finite-dimensional space (Thm. 35.19, p. 313), the stride acts on it (Thm. 35.3, p. 308), and the relations that present it are the letter and mark relations of the period (Thm. 8.3, p. 54) read at the data of the ray. What is new is how to use them: finitely many of these relations, on a box of insertions measured by the walls, already reduce the stride to a matrix on a set of masters, and the law is that matrix read through one vector. Each statement below is proved for a ray through the origin of one list strictly inside its walls, and holds for a ray with an origin with the changes of Remark 23.13; a family of tables changes only the notation.

The construction. Insertions at the levels $(a,b) \in \{0,1\}^2$ of the two letters, of wall degree at most T (Definition 23.1); the letter identities among them and the mark relations of the Laurent insertions (Proposition 23.3); one elimination. What is left of the Laurent insertions of wall degree below T are the masters $b_0 = 1, b_1, \dots, b_{\mu-1}$, and the stride acts on them by the *one-step matrix*,

$$\langle b_j \rangle_{k-1} = \sum_l B_{jl}(k)\, \langle b_l \rangle_k \qquad (B \in \mathbb{Q}(k)^{\mu\times\mu}),$$

an identity of meromorphic functions of k (Theorem 23.5). The law of $\langle 1\rangle_k$ is the first dependence among the rows $e_0 B(k-i+1)\cdots B(k)$, of order at most μ, and its certificate is the record of the elimination: Laurent insertions $G_{j,e}$ with $Lb_j - \sum_l B_{jl} b_l = \sum_e (k\, c_e + \partial_{y_e}) G_{j,e}$. The native values are the spectrum of B at infinite strength, with the masters' values at the native gauges as eigenvectors, when B is finite there or is made so by a change of masters (Proposition 23.11). $T = 2$ or 3 in every case met, from three marks to eight.

§23.1 sets up the box and its relations; §23.2 shows on the three-mark centre why the torus alone does not close, with the extra solution named; §23.3 is the theorem, and the theorem that a search through the boxes stops; §23.4 the native values; §23.5 the procedure, which replaces step (3) of Procedure 18.6 and carries the reading through to the value, and a program that runs it from the list and the direction alone; §23.6 the receipts, from the three-mark centre, worked by hand, to V at both directions; §23.7 what it shows. The receipts are in `receipts_s23.zip`, and the program in `rayreader.zip` (python-flint, numpy, mpmath and a C eliminator for sparse linear algebra modulo a prime; no quadrature and no closing enters a construction, and closings enter only the checks).

Throughout, $(\mathcal{E}, \mathcal{B})$ is a list of rank r on $n = |\mathcal{E}|$ marks with letters K and Ψ, $(\mathbf{m}_1; s, t)$ is an integral direction on the balance wall with $s, t \in \mathbb{N}$ and $\kappa_{1,S} > 0$ for every nonempty proper S, and

$$f = \alpha^{\mathbf{m}_1} K^{-s} \Psi^{-t}, \qquad L = \frac{1}{f} = \alpha^{-\mathbf{m}_1} K^{s} \Psi^{t}, \qquad \langle g \rangle_k = \int_{y_{e_0}=0} g\, f^k\, dy,$$

the stride, its reciprocal (a Laurent polynomial), and the values (Def. 35.1, p. 307). For $S \subseteq \mathcal{E}$, $r(S)$ is the largest trace of a basis on S and $r^*(S) = |S| - r + r(\mathcal{E} \setminus S)$ the largest trace of a cobasis (Def. 7.10, p. 49). The defects of the direction are $c_e = \partial_{y_e} \log f = m_{1,e} - sR_e - tR_e^*$, with the occupations $R_e = \sum_{T \ni e} \alpha^T / K$ and $R_e^* = \sum_{T \not\ni e} \alpha^{\mathcal{E}\setminus T} / \Psi$ (Def. 35.1; Lemma 35.2(ii), p. 308).

23.1 The box of the walls

Definition 23.1 (Levels, wall degree, the box)**.** An *insertion at the level* $(a,b) \in \mathbb{N}^2$ is

$$g = \alpha^u K^{-a} \Psi^{-b}, \qquad u \in \mathbb{Z}^{\mathcal{E}}, \quad |u| = a\, r + b\,(n - r),$$

a regular function on the letter complement of degree 0; products of monomials and chances (Def. 35.1) are finite sums of these, so they span the insertions. Its value is a period at data on the line of the ray,

$$\langle g \rangle_k = P_{\mathcal{B}}\big(k\mathbf{m}_1 + u;\ ks + a,\ kt + b\big)$$

(Lemma 35.2(iii), p. 308). Its *wall degree* is

$$\deg g = \max_{\varnothing \neq S \subsetneq \mathcal{E}} \frac{u(S) - a\, r(S) - b\, r^*(S)}{\kappa_{1,\mathcal{E}\setminus S}}.$$

For $T \in \mathbb{N}$ the *box* $\mathfrak{B}_T$ is the set of insertions of wall degree at most T at the levels (a,b) with $a \le [s > 0]$ and $b \le [t > 0]$: one unit of each active letter. Insertions are taken modulo the

automorphisms of the list that fix $\mathbf{m}_1$, and at the centre ($s = t$, $\mathbf{m}_1 = s\mathbf{1}$) also modulo Cremona, $\alpha^u K^{-a}\Psi^{-b} \mapsto \alpha^{-u+(a+b)\mathbf{1}} K^{-b}\Psi^{-a}$ (Thm. 8.13, p. 57); both fix every value.

Lemma 23.2 (The wall degree)**.** [derived]

(i) $\kappa_{1,\mathcal{E}\setminus S} = s\,r(S) + t\,r^*(S) - m_1(S)$*, the growth of* L *when the marks of* S *grow together; so* $\deg L = 1$*, and* $\deg(Lg) = \deg g + 1$ *for every insertion* g*.*

(ii) On the chart, $|g| \le C\, f^{-\deg g}$*, and no smaller power of* f^{-1} *bounds* g*.*

(iii) $\deg(\partial_{y_e} g) \le \deg g$*,* $\deg(R_e g) \le \deg g$ *and* $\deg(R_e^* g) \le \deg g$ *termwise, and every term of the letter identities of* g *(Proposition 23.3(i)) has wall degree at most* $\deg g$*.*

(iv) At each level the insertions of wall degree at most T *are the lattice points of a polytope dilated by* T*; there are finitely many,* $O(T^{n-1})$*.*

Proof. (i) By Def. 8.1, p. 53, $\kappa_{1,X} = m_1(X) - s\,d(X) - t\,d^*(X)$ with $d(X)$ the least trace of a basis and $d^*(X) = |X| - r(X)$ the least trace of a cobasis. For $X = \mathcal{E}\setminus S$, $d(X) = r - r(S)$ and $d^*(X) = n - r - r^*(S)$, and the balance $m_{1,\mathcal{E}} = sr + t(n-r)$ gives the formula. Along $\alpha = \lambda^{\mathbf{1}_S}\beta$, $\lambda \to \infty$, the largest power of λ in K is $\lambda^{r(S)}$, in Ψ it is $\lambda^{r^*(S)}$, so L grows like $\lambda^{\kappa_{1,\mathcal{E}\setminus S}}$ and Lg adds exactly $\kappa_{1,\mathcal{E}\setminus S}$ to the numerator of every quotient. (ii) On the sector $\alpha_{\pi(1)} \ge \cdots \ge \alpha_{\pi(n)}$ one basis dominates every monomial of K and one cobasis every monomial of Ψ (Lemma 8.5, p. 55), so $\log|g|$ and $\log f$ are, up to bounded terms, linear forms in y on the sector. Modulo $\mathbf{1}$ the sector is the cone spanned by the indicators of its heads $\{\pi(1),\dots,\pi(i)\}$, $i < n$, along which the two forms are $u(S) - a\,r(S) - b\,r^*(S)$ and $-\kappa_{1,\mathcal{E}\setminus S} < 0$, the dominating basis being at the top of every head. A linear inequality holds on a simplicial cone exactly when it holds on its generators. (iii) $\partial_{y_e} g = (u_e - aR_e - bR_e^*)g$ with $0 \le R_e, R_e^* \le 1$, the chances that e lies in a basis and in a cobasis; each term of a sum of positive functions is at most the sum on the orthant, and an insertion bounded on the orthant by a positive multiple of g has wall degree at most $\deg g$, by (ii). (iv) Each quotient bounds $u(S)$ by an affine function of T, and the bounds over all S, with $|u|$ fixed, cut out a polytope of the hyperplane $|u| = ar + b(n-r)$, bounded because the walls are positive. □

So a box is the set of insertions dominated by a power of the reciprocal stride, and the stride moves the wall degree by exactly one: L carries an insertion of degree at most $T-1$ into the box $\mathfrak{B}_T$.

Proposition 23.3 (The relations on a box are those of the period)**.** [derived] *For an insertion* $g = \alpha^u K^{-a}\Psi^{-b}$ *and a mark* e*, as identities of meromorphic functions of* k*:*

(i) (letter identities) $\langle g\rangle_k = \sum_{T\in\mathcal{B}} \langle \alpha^{\mathbf{1}_T} g K^{-1}\rangle_k = \sum_{T\in\mathcal{B}} \langle \alpha^{\mathbf{1}_{\mathcal{E}\setminus T}} g \Psi^{-1}\rangle_k$*;*

(ii) (mark relations) $\big\langle (k\,c_e + \partial_{y_e}) g\big\rangle_k = 0$*, that is*

$$(k\,m_{1,e} + u_e)\,\langle g\rangle_k = (ks + a)\sum_{T\ni e}\big\langle \alpha^{\mathbf{1}_T} g K^{-1}\big\rangle_k + (kt + b)\sum_{T\not\ni e}\big\langle \alpha^{\mathbf{1}_{\mathcal{E}\setminus T}} g\Psi^{-1}\big\rangle_k;$$

(iii) (the stride) $\langle g\rangle_{k-1} = \langle Lg\rangle_k$*.*

Proof. With $\langle \alpha^v K^{-a'}\Psi^{-b'}\rangle_k = P_{\mathcal{B}}(k\mathbf{m}_1 + v; ks + a', kt + b')$, (i) is the letter relation of Thm. 8.3(i), p. 54, and (ii) is its mark relation, Thm. 8.3(ii)–(iii), at the data $(k\mathbf{m}_1 + u; ks + a, kt + b)$: on the balance wall the pure mass shift carries the factor $\kappa_{\mathcal{E}} = 0$, and $\sum_{T\ni e}\alpha^T = \partial_{y_e} K$, $\sum_{T\not\ni e}\alpha^{\mathcal{E}\setminus T} = \partial_{y_e}\Psi$. Thm. 8.3 holds as identities of meromorphic functions of the data, and the line $k \mapsto (k\mathbf{m}_1 + u; ks + a, kt + b)$ meets each pole hyperplane $\kappa_S = -j$ in at most one point, because its walls are $k\kappa_{1,S} + \kappa_S(u; a, b)$ with $\kappa_{1,S} \ne 0$; so the restrictions are meromorphic in k and the identities restrict. The second form of (ii) is the first written out, since $\partial_{y_e} g = (u_e - aR_e - bR_e^*)g$ and $c_e = m_{1,e} - sR_e - tR_e^*$: it is the relation of Thm. 35.3, p. 308, with the origin 0. (iii) is $Lgf^k = gf^{k-1}$. □

The relations of the box. $\mathcal{R}_T$ consists of the letter identities (i) of the insertions of $\mathfrak{B}_T$ at the levels with $a = 0$ (for K, when $s > 0$) and with $b = 0$ (for Ψ, when $t > 0$), and the mark relations (ii) of the *seeds*, the Laurent insertions α^v of wall degree at most T, at every mark. By Lemma 23.2(iii) every term of every relation of $\mathcal{R}_T$ lies in $\mathfrak{B}_T$: the mark relation of a seed α^v has its terms at the levels $(0,0)$, $(1,0)$ and $(0,1)$, the occupations each adding one unit of one letter, and the letter identities tie those levels to $(1,1)$. In words: *the mark relations of the Laurent seeds, with one unit of each letter exponent for the occupations to land in.*

23.2 The torus alone does not close

The closings are values on the second chain (Remark 18.3, Proposition 22.2), and the relations they obey among themselves are the mark relations of the insertions $\alpha^w L$, which are Laurent polynomials. Since $c_e = -\partial_{y_e} L/L$,

$$(k\,c_e + \partial_{y_e})(\alpha^w L) = \alpha^w\big(w_e\,L + (1-k)\,\partial_{y_e} L\big) = \sum_v \lambda_v\,\big(w_e + (1-k)v_e\big)\,\alpha^{w+v}, \qquad L = \sum_v \lambda_v \alpha^v.$$

Call these the *closing relations*: for the unknowns $X(w) = \langle \alpha^w \rangle_k$ they read $\sum_v \lambda_v (w_e + (1-k)v_e) X(w+v) = 0$. They are the relations that a recurrence found from closings can lean on, and the first thing to run is whether they reduce the stride.

Proposition 23.4 (The closing relations have extra solutions)**.** [derived]*; the counts* [computed]*.*

(i) Let $z \in (\mathbb{C}^\times)^{\mathcal{E}}$ be a point at which L and every $\partial_{y_e} L$ vanish, a singular point of the zero set of L on the torus. Then $X(w) = z^w$ solves every closing relation, at every k.

(ii) At the three-mark centre $L = K\Psi/\alpha^{\mathbf{1}} = e_1 e_2/e_3$, and $z = (1, \omega, \omega^2)$, $\omega = e^{2\pi i/3}$, has $K(z) = \Psi(z) = 0$, so (i) applies. Under the six permutations and Cremona its orbit is $\{(1,\omega,\omega^2), (1,\omega^2,\omega)\}$ up to scale.

(iii) On the Laurent insertions of wall degree at most R, modulo the symmetries, for $R = 3, 4, 5, 6, 7$, the closing relations leave three masters,

$$1, \qquad b_1 = \tfrac{1}{6}\sum_{i\neq j} \frac{\alpha_i}{\alpha_j}, \qquad b_2 = \tfrac{1}{6}\sum_{i\neq j} \frac{\alpha_i^2}{\alpha_j^2},$$

and the left side of the law (L) of §19 at $k = 7$, read at the strength 9 as $49\langle L^2\rangle_9 - 563\langle L\rangle_9 + 576\langle 1\rangle_9$, reduces to

$$504 + 248\,b_1 - 392\,b_2 \;\neq 0;$$

at $k = 10$ and $k = \frac{23}{3}$ the residuals are $880 + \frac{946}{3}b_1 - \frac{2200}{3}b_2$ and $\frac{8671}{15} + \frac{1313}{5}b_1 - \frac{6877}{15}b_2$. On the extra solution of (ii), where $L = 0$ and $b_1 = b_2 = -\frac{1}{2}$, each residual is $9(k+1)^2$ (576, 1089, 676): the first two terms of (L) vanish with L, and the last, $9(k+1)^2\langle 1\rangle_{k+2}$, does not.

(iv) In the box $\mathfrak{B}_2$ of Definition 23.1, which adds one unit of each letter exponent, the masters are 1 and b_1, and (L) follows (Example 23.15).

Proof. (i) $\sum_v \lambda_v (w_e + (1-k)v_e) z^{w+v} = z^w\big(w_e L(z) + (1-k)\,\partial_{y_e} L(z)\big) = 0$. (ii) $e_1(z) = 1+\omega+\omega^2 = 0$ and $e_2(z) = \omega+\omega^2+\omega^3 = 0$, so L and its first derivatives, each a sum of products with a factor e_1 or e_2, vanish; the permutations and $\alpha \mapsto \alpha^{-1}$ carry z to its images and to $(1,\omega^2,\omega)$. At z the six ratios α_i/α_j are three ω and three ω^2, and so are their squares, so $b_1(z) = b_2(z) = \frac{1}{6}(3\omega + 3\omega^2) = -\frac{1}{2}$. (iii) and (iv) are exact computations over $\mathbb{Q}$, at the strengths $7, 10, \frac{23}{3}$ for (iii) and $5, 9, \frac{17}{2}$ for (iv). $\square$

So the closing relations cannot prove (L): they have a solution on which (L) fails. A recurrence read off closings is a recurrence of one solution of the closing relations, and those relations have more solutions than the ray. The extra ones sit where the letters vanish on the torus. A letter identity, $\alpha^u = \sum_T \alpha^{u+\mathbf{1}_T} K^{-1}$, has no solution supported at a zero of K, because an insertion with K^{-1} has no value there; one unit of each letter exponent is what removes them, and in every case met it is all that is needed (§23.6).

23.3 The masters and the one-step matrix

Order the insertions of $\mathfrak{B}_T$ by level, then by wall degree, higher first, and eliminate the relations $\mathcal{R}_T$ over $\mathbb{Q}(k)$ in that order. The Laurent insertions of wall degree at most $T-1$ that are not eliminated are the *masters*, $b_0 = 1, b_1, \dots, b_{\mu-1}$; by Lemma 23.2(i) each Lb_j, expanded into Laurent monomials, lies in $\mathfrak{B}_T$, and it is a *target*. Let $\mathcal{L}_k$ be the relation space of the ray at the strength k, the $\mathbb{C}(k)$-span of the $(k\,c_e + \partial_{y_e})g$ (Def. 35.4, p. 308).

Theorem 23.5 (The stride on the masters)**.** [derived] *Suppose that every target reduces to the masters modulo $\mathcal{R}_T$:*

$$Lb_j \equiv \sum_{l<\mu} B_{jl}(k)\, b_l \pmod{\mathcal{R}_T}, \qquad B_{jl} \in \mathbb{Q}(k).$$

Put $V(k) = (\langle b_l \rangle_k)_{l<\mu}$; each master, and each insertion of $\mathfrak{B}_T$, stands for the average over its orbit under the symmetries.

*(i) (*The one-step law*) $V(k-1) = B(k)\,V(k)$, as an identity of meromorphic functions of k.*

*(ii) (*The certificate*) There are Laurent polynomials $G_{j,e}$ with coefficients in $\mathbb{Q}(k)$, supported on the seeds, so of wall degree at most T, with*

$$Lb_j - \sum_l B_{jl}(k)\, b_l = \sum_{e\in\mathcal{E}} \big(k\,c_e + \partial_{y_e}\big) G_{j,e}$$

as an identity of rational functions of α and k; it is a certificate in the sense of §19, Step 4, for the one-step law.

*(iii) (*The law*) Put $w_0 = (1, 0, \dots, 0)$ and $w_i(k) = w_0\, B(k-i+1)\cdots B(k)$. Then $\langle 1 \rangle_{k-i} = w_i(k)\,V(k)$. Let h be the least index at which $w_0, \dots, w_h$ are linearly dependent over $\mathbb{Q}(k)$, so $h \le \mu$, and $\sum_{i\le h} q_i(k)\, w_i(k) = 0$ with coprime $q_i \in \mathbb{Q}[k]$. With $p_j(k) = q_{h-j}(k+h)$,*

$$\sum_{j=0}^{h} p_j(k)\, \langle 1 \rangle_{k+j} = 0$$

identically in k, and $\sum_j p_j(k)\, f^j \in \mathcal{L}_k$: it is a law of the ray in the sense of Cor. 35.6, p. 308.

(iv) The same holds for every insertion of $\mathfrak{B}_T$ that reduces to the masters, with its own row in place of w_0.

Proof. (i) The reduction is an identity, in the space of formal combinations of insertions of $\mathfrak{B}_T$ over $\mathbb{Q}(k)$, between $Lb_j - \sum_l B_{jl} b_l$ and a combination of elements of $\mathcal{R}_T$. By Proposition 23.3(i)–(ii) every element of $\mathcal{R}_T$ has the value 0 identically in k, and identifying an insertion with its images under the symmetries does not change values; so $\langle Lb_j \rangle_k = \sum_l B_{jl}(k) \langle b_l \rangle_k$, and $\langle Lb_j \rangle_k = \langle b_j \rangle_{k-1}$ by Proposition 23.3(iii). (ii) The letter identities are identities of functions, so read as functions the combination keeps only the mark relations of the seeds; for each mark e, $G_{j,e}$ is the $\mathbb{Q}(k)$-combination of the seeds α^v whose mark relation at e the reduction used (an orbit representative is replaced by

the average over its orbit, which is again supported on seeds). Its wall degree is at most T by the definition of the seeds. (iii) $\langle 1\rangle_{k-1} = \langle L\rangle_k$, and inductively, by (i), $\langle 1\rangle_{k-i} = w_0 B(k-i+1)\langle b\rangle_{k-i+1} = \cdots = w_i(k)V(k)$. The w_i lie in $\mathbb{Q}(k)^\mu$, so $h \le \mu$. For the membership: by (ii), $L^i \equiv \sum_l w_{i,l}(k)\, b_l$ modulo $\mathcal{L}_k$, since $L\mathcal{L}_k = \mathcal{L}_{k+1}$, from $L\,(k\,c_e + \partial_{y_e})g = ((k+1)\,c_e + \partial_{y_e})(Lg)$, carries the reduction at one strength to the next. So $\sum_i q_i(k)L^i \in \mathcal{L}_k$, and multiplying by f^h, which carries $\mathcal{L}_k$ to $\mathcal{L}_{k-h}$, $\sum_i q_i(k) f^{h-i} \in \mathcal{L}_{k-h}$; at $k \mapsto k+h$ this is $\sum_j p_j(k) f^j \in \mathcal{L}_k$. The identity in k follows from (i). (iv) is (iii) with the row of the insertion. □

Corollary 23.6 (The law at every strength)**.** [derived] *The law of Theorem 23.5(iii) holds as an identity of meromorphic functions of k, so at every real $k > 0$, where all its terms converge, and as $k \to 0^+$ in Corollary 18.2: the corner relation $[k^{n-1}]p_0 \cdot \mathrm{Cor} + \sum_{j\ge 1} p_j(0)P(j) = 0$ and the recurrence $\sum_j p_j(-i)\, x_{i-j} = 0$ of the closings follow. This is §19, Step 5, for every law the construction produces, with no bounds on the pieces: the identity in k comes from Thm. 8.3, p. 54, through Proposition 23.3.*

Proof. Theorem 23.5(iii) gives the identity; at $\mathrm{Re}\, k > 0$ every $\langle 1\rangle_{k+j}$ converges (Thm. 35.10(i), p. 309), so the identity of meromorphic functions is an identity of the convergent integrals there. Corollary 18.2 needs exactly the law as an identity of meromorphic functions. □

Remark 23.7 (The certificate also integrates directly). The route of §19, Step 5, is available as well, with its bounds read off the box. By Lemma 23.2(ii)–(iii), $|G_{j,e}f^k| \le C f^{k-T}$ and $|\partial_{y_e}(G_{j,e}f^k)| \le C(1+|k|)f^{k-T}$ on the chart, and $f \le C' e^{-c'\|y\|}$ (Thm. 35.10(i)); so for real $k > T$ the boundary terms of the integration by parts vanish, the mark e_0 of the chart being handled by Euler's identity as in §19, Step 5(b), and (ii) integrates to (i) there. The wall degree of the box is the strength from which the certificate integrates, and the continuation in k does the rest.

The search. Whether a box makes every target reduce is checked, not assumed. That some box does follows from the finite dimension of the ray, once the levels are a parameter too.

Definition 23.8 (Boxes with levels)**.** Let $T, A, B \in \mathbb{N}$ with $A \ge [s > 0]$ and $B \ge [t > 0]$, and put $A^- = \max(A-1, 0)$, $B^- = \max(B-1, 0)$. The box $\mathfrak{B}_T^{A,B}$ is the set of insertions of wall degree at most T at the levels (a, b) with $a \le A$ and $b \le B$. Its relations $\mathcal{R}_T^{A,B}$ are the letter identities of its insertions at the levels with $a < A$ (for K) and with $b < B$ (for Ψ), and the mark relations, at every mark, of its *seeds*, its insertions at the level (A^-, B^-). So $\mathfrak{B}_T = \mathfrak{B}_T^{[s>0],[t>0]}$, with the Laurent seeds of Definition 23.1.

Every term of every relation of $\mathcal{R}_T^{A,B}$ lies in $\mathfrak{B}_T^{A,B}$, by Lemma 23.2(iii): the mark relation of a seed has its terms at (A^-, B^-), at $(A^- + 1, B^-)$ with the factor $ks + A^-$, and at $(A^-, B^- + 1)$ with the factor $kt + B^-$, and a factor that is not identically zero comes with a level inside the box.

Lemma 23.9 (Combinations that vanish as functions)**.** [derived]

(i) A combination over $\mathbb{Q}(k)$ of insertions of $\mathfrak{B}_T^{A,B}$, taken as formal symbols, that is zero as a function of α is a combination of the letter identities of $\mathcal{R}_T^{A,B}$.

(ii) For every insertion g of wall degree at most T at a level $(a, b) \le (A^-, B^-)$ and every mark e, the mark relation of g is a combination of $\mathcal{R}_T^{A,B}$.

Both hold for the quotient of the box by the symmetries, insertions identified with their images and relations with their averages over the orbits, when the symmetries carry the box to itself: always for the automorphisms of the list that fix $\mathbf{m}_1$, which keep levels and wall degrees, and for Cremona when $A = B$, since at the centre $\kappa_{1,S} = \kappa_{1,\mathcal{E}\setminus S}$ and Cremona keeps the wall degree and exchanges the two levels.

Proof. (i) A letter identity replaces an insertion at the level (a, b) by insertions at $(a+1, b)$, or at $(a, b+1)$, of no larger wall degree (Lemma 23.2(iii)). Applying them, every term of the combination is carried to the level (A, B) inside the box, where the insertions are the $\alpha^u K^{-A}\Psi^{-B}$ with distinct u; such a combination is zero as a function only if its coefficients are, since $K^A\Psi^B$ times it is a Laurent polynomial. So the combination is the sum of the letter identities applied. (ii) Multiply $g = \alpha^u K^{-a}\Psi^{-b}$ above and below by $K^{A^- - a}\Psi^{B^- - b} = \sum_v \lambda_v \alpha^v$: as functions $g = \sum_v \lambda_v\, \alpha^{u+v} K^{-A^-}\Psi^{-B^-}$, and the insertions on the right are seeds, of wall degree at most $\deg g$, since each v has $v(S) \le (A^- - a)\, r(S) + (B^- - b)\, r^*(S)$. The mark relation of g less the same combination of the mark relations of these seeds is zero as a function, because $k\, c_e + \partial_{y_e}$ is linear; its terms lie in the box, by Lemma 23.2(iii) and because a term at the level $a+1$ (or $b+1$) has the factor $ks + a$ (or $kt + b$), which is zero when $a = A = 0$ (or $b = B = 0$), as then $s = 0$ (or $t = 0$). So (i) applies. For the quotient, write a combination of classes as the combination of the averages over their orbits, apply (i) or (ii) to it, and identify each insertion with its class: the identification is linear, and carries a relation of a box that the symmetries carry to itself to a relation of its quotient. □

Theorem 23.10 (The search stops)**.** [derived] *For $T \ge 1$, A, B as in Definition 23.8 and $0 \le D < T$, call* (T, A, B, D) accepted *when the masters, the Laurent insertions of wall degree at most D that the elimination of $\mathcal{R}_T^{A,B}$ over $\mathbb{Q}(k)$ leaves free, are such that every target Lb lies in $\mathfrak{B}_T^{A,B}$ and reduces to them. Then some (T, A, B, D) is accepted, so a search that runs through all of them, in any order that reaches each, stops. With the symmetries the same holds, with $A = B$ when Cremona is among them.*

Proof. Let $\mathcal{Q} = (\mathbb{C}[X_F] \otimes \mathbb{C}(k))/\mathcal{L}$, the space whose dimension is the dimension of the ray, finite by Thm. 35.19(i), p. 313 ($\mathcal{L}$ the relation space of Def. 35.4), and $\mathcal{Q}_0 \subseteq \mathcal{Q}$ the span of the classes of the Laurent insertions. The classes of the Laurent insertions of wall degree at most D, $D = 0, 1, 2, \ldots$, span an increasing chain of subspaces of $\mathcal{Q}_0$ whose union is $\mathcal{Q}_0$, so it is constant from some D_0 on. The Laurent insertions $g_1, \ldots, g_N$ of wall degree at most $D_0 + 1$ are finitely many (Lemma 23.2(iv)), and for each there are Laurent insertions h of wall degree at most D_0 with

$$g_i - \sum_h \lambda_{i,h}\, h = \sum_{j,e} \nu_{i,j,e} \left(k\, c_e + \partial_{y_e}\right) G_{i,j}$$

as functions, for finitely many insertions $G_{i,j}$ and $\lambda, \nu \in \mathbb{C}(k)$; as the relations have rational coefficients, λ and ν may be taken in $\mathbb{Q}(k)$. Take $T > D_0$ at least the wall degree of every $G_{i,j}$, and A, B with every $G_{i,j}$ at a level at most (A^-, B^-). By Lemma 23.9(ii) the mark relations of the $G_{i,j}$ are combinations of $\mathcal{R}_T^{A,B}$, and then so is the left side above, by Lemma 23.9(i) applied to its difference with them: $g_i \equiv \sum_h \lambda_{i,h} h$ modulo $\mathcal{R}_T^{A,B}$.

The elimination orders the insertions by total level $a + b$ and then by wall degree, so a relation whose most complex term is Laurent has only Laurent terms of no larger wall degree, and the normal form of a Laurent insertion h is a combination of free Laurent insertions of wall degree at most $\deg h$. Take $D = D_0$. Every monomial of a target Lb, b a master, has wall degree at most $\deg b + 1 \le D_0 + 1$, since for a monomial α^w of $K^s\Psi^t$ one has $w(S) - m_1(S) \le \kappa_{1,\mathcal{E}\setminus S}$ (Lemma 23.2(i)); so the target is a combination of the g_i, in the box, and its normal form is a combination of the normal forms of insertions of wall degree at most D_0, that is, of masters. With the symmetries, identifying each insertion with its class carries the congruences $g_i \equiv \sum_h \lambda_{i,h} h$ to congruences modulo the relations of the quotient (Lemma 23.9, last part), and the rest is the same, an average of monomials having the wall degree of each of them. □

So the construction is finite and sound. The box is finite (Lemma 23.2(iv)); its relations are exactly those of the period (Proposition 23.3), so nothing beyond Thm. 8.3 is used and every step

is an identity, so that whatever the box, a law it produces is a law of the ray; and a search through the boxes stops (Theorem 23.10). The proof bounds neither T nor the levels; in every case met, a box with one unit of each active letter and $T = 2$ or 3 is accepted, with $D = T - 1$. Minimality is a separate matter. The masters span a space that contains 1 and is closed under the stride, which is all the law needs; when they are independent modulo $\mathcal{L}_k$ (Proposition 23.11(ii) is a test), the order h is the dimension of the cyclic space of 1 in it, and no shorter law follows from the relations.

23.4 The native values are the spectrum at infinite strength

Proposition 23.11 (The spectrum at infinite strength)**.** [derived] *In the setting of Theorem 23.5, suppose in (i) and (ii) that $B(k)$ has a finite limit $B(\infty)$ as $k \to \infty$.*

(i) For every nondegenerate native gauge α_c of the ray (Def. 35.4, p. 308) the vector $\beta_c = (b_l(\alpha_c))_{l<\mu}$ of the masters' values there satisfies

$$B(\infty)\,\beta_c = L(\alpha_c)\,\beta_c, \qquad \beta_c \neq 0 :$$

the reciprocal $L(\alpha_c) = f(\alpha_c)^{-1}$ of every native value is an eigenvalue of $B(\infty)$, with the masters' values at the gauge as an eigenvector.

(ii) If the vectors β_c of some nondegenerate native gauges span $\mathbb{C}^\mu$, the masters are linearly independent modulo $\mathcal{L}_k$, $B(\infty)$ is diagonalizable, and its eigenvalues are exactly the reciprocals of the native values of those gauges.

(iii) If $B(k)$ has no finite limit but a change of masters $\tilde{b} = T(k)\,b$, $T \in \mathrm{GL}_\mu(\mathbb{Q}(k))$, gives

$$\tilde{B}(k) = T(k-1)\,B(k)\,T(k)^{-1}$$

a finite limit $\tilde{B}(\infty)$, then the reciprocal $L(\alpha_c)$ of every nondegenerate native value is an eigenvalue of $\tilde{B}(\infty)$, with the leading coefficient of $T(k)A^{(c)}(b)$ (Thm. 35.5, as in the proof) as an eigenvector. If moreover $T(k-1)T(k)^{-1} \to 1$, the eigenvalues of $B(k)$ itself tend to those of $\tilde{B}(\infty)$.

Proof. (i) By Thm. 35.5(ii), p. 308, the map $A^{(c)}$, which sends an insertion to the asymptotic series of its value over a steepest-descent disc through α_c, vanishes on $\mathcal{L}_k$ and is $\mathbb{C}(k)$-linear. Applied to Theorem 23.5(ii),

$$A^{(c)}(Lb_j) = \sum_l B_{jl}(k)\,A^{(c)}(b_l) \qquad \text{in } \mathbb{C}((1/k)).$$

By Thm. 35.5(i) the leading coefficient of $A^{(c)}(g)$ is $(2\pi)^{(n-1)/2}(\omega_0/dv)(\alpha_c)\,g(\alpha_c)$, the same nonzero factor for every insertion g, all of order $f(\alpha_c)^k k^{-(n-1)/2}$. Comparing leading coefficients, with $B(k) = B(\infty) + O(1/k)$, gives $L(\alpha_c)b_j(\alpha_c) = \sum_l B_{jl}(\infty)b_l(\alpha_c)$. And $b_0 = 1$, so $\beta_c \neq 0$. (ii) A relation $\sum_l \lambda_l(k)b_l \in \mathcal{L}_k$ with $\lambda \neq 0$, scaled so that $\lambda(\infty)$ is finite and nonzero, gives by the same comparison $\sum_l \lambda_l(\infty)b_l(\alpha_c) = 0$ for every c, so $\lambda(\infty)$ is orthogonal to a spanning set: a contradiction. The β_c are eigenvectors by (i), and μ of them are independent. (iii) The disc through α_c does not depend on k, and $\langle Lg\rangle_k = \langle g\rangle_{k-1}$ over it as on the cycle, so $A^{(c)}(Lg)(k) = A^{(c)}(g)(k-1)$, and Theorem 23.5(ii) gives $Y(k-1) = \tilde{B}(k)\,Y(k)$ for $Y(k) = T(k)A^{(c)}(b)(k)$. The entry $A^{(c)}(b_0)$ has a nonzero leading coefficient and $T(k)$ is invertible, so $Y(k) = F(k)\,k^e\big(\tilde{\beta}_c + O(1/k)\big)$ with $\tilde{\beta}_c \neq 0$, some $e \in \mathbb{Z}$, and $F(k) = f(\alpha_c)^k k^{-(n-1)/2}$; and $F(k-1)/F(k) = L(\alpha_c)\big(1 + O(1/k)\big)$. Dividing by $F(k)k^e$ and letting $k \to \infty$ gives $L(\alpha_c)\tilde{\beta}_c = \tilde{B}(\infty)\tilde{\beta}_c$. Finally $T(k)B(k)T(k)^{-1} = \big(T(k-1)T(k)^{-1}\big)^{-1}\tilde{B}(k)$, which tends to $\tilde{B}(\infty)$ when $T(k-1)T(k)^{-1} \to 1$, and its eigenvalues are those of $B(k)$. □

So a census of native gauges, which Proposition 20.37(iii) could only bound from below, is bounded from above by one matrix: the characteristic polynomial of $B(\infty)$ contains the reciprocal of every nondegenerate native value, with the masters at the gauge as its eigenvector, and when those eigenvectors span it is exactly that list. The symbol of the law of Theorem 23.5(iii) vanishes at the native values (Cor. 35.6), which are among the reciprocal eigenvalues; Table 23.16 compares the two on every ray. When the masters take proportional values at two native gauges of different native values, (i) shows that $B(\infty)$ cannot be finite. On the fixture (Theorem 20.15) the masters are 1 and $b_1 = \alpha^{T_\triangle}/\alpha^{T_\star}$, the triangle's three marks over the star's, and at both native gauges, of values $\frac{1}{256}$ and $\frac{1}{4}$, $b_1 = 1$ [computed] (Newton's method from random complex starts finds no other native value); so $\beta_c = (1, 1)$ for two different eigenvalues, and indeed $B_{01}(k) = \frac{28k^3-14k^2}{k^2-k+1}$ grows. The master $k(b_1 - b_0)$ in place of b_1 repairs it: with $T(k) = \left(\begin{smallmatrix} 1 & 0 \\ -k & k \end{smallmatrix}\right)$ the matrix $\tilde{B}$ of (iii) has the limit $\left(\begin{smallmatrix} 88 & 28 \\ 504 & 172 \end{smallmatrix}\right)$, of eigenvalues 256 and 4 [computed]; and $T(k-1)T(k)^{-1} = \operatorname{diag}\bigl(1, \frac{k-1}{k}\bigr) \to 1$, so by (iii) the eigenvalues of $B(k)$ itself tend to 256 and 4, as they do: its trace tends to 260 and its determinant to 1024 [computed].

Without the condition $T(k-1)T(k)^{-1} \to 1$ the limits of the eigenvalues of a matrix that is not finite at infinity say nothing. For B constant and $T(k) = \left(\begin{smallmatrix} 1 & 0 \\ k & 1 \end{smallmatrix}\right)$, the matrix $T(k-1)BT(k)^{-1}$ has the trace $\operatorname{tr} B - B_{01}$ and the determinant $\det B$ at every k, while the values $T(k)B^{-k}v$ of its system go like λ^{-k} times a polynomial in k, for the eigenvalues λ of B; here $T(k-1)T(k)^{-1} = \left(\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right)$. So when $B(k)$ grows, its spectrum is a census only through a change of masters as in (iii), and V at its lean (Proposition 23.18(iv)) is the case where none is known.

23.5 The procedure, and a program that runs it

Procedure 23.12 (A ray read from its list and its direction)**.** Input: a list, a direction strictly inside its walls, and the number of digits wanted. Output: the masters, the one-step matrix B over $\mathbb{Q}(k)$, the law of $\langle 1 \rangle_k$ over $\mathbb{Z}$, and the value $\langle 1 \rangle_1 = P_{\mathcal{B}}(\mathbf{m}_1; s, t)$. Steps (1)–(4) replace step (3) of Procedure 18.6, and step (5) is its reading by the gauge end.

(1) *The walls and the symmetries.* The walls $\kappa_{1,S}$ (Def. 8.1), the automorphisms of the list that fix $\mathbf{m}_1$, and Cremona at the centre. Rational masses are read through an origin (Remark 23.13).

(2) *The masters.* The boxes $\mathfrak{B}_T^{A,B}$ in order of $(T-2) + (A - [s > 0]) + (B - [t > 0])$, larger T first. In each, one elimination modulo a prime at a random strength, the insertions ordered by total level, then by wall degree, then by a fixed order of exponents; for $D = T-1, T-2, \ldots, 0$ the masters of wall degree at most D, and one more elimination, with their targets. The first (T, A, B, D) at which every target lies in the box and reduces to the masters is taken. The search stops (Theorem 23.10).

(3) *The one-step matrix over* $\mathbb{Q}(k)$*.*

  (a) *The relations the targets need.* At two random strengths, the relations whose pivot rows enter the reduction of a target, directly or through the reduction of another such relation, are recorded; the others are dropped. Every set of relations between these and the whole box gives the targets the same normal forms: a leading column of a subspace of the row space is a leading column of the whole, so the masters are free for every such set, and a normal form is unique given the row space and its free columns.

  (b) *Modulo primes.* Primes $q \in (2^{61}, 2^{62})$ and strengths drawn at random. On the first prime, as many strengths as determine every entry of B as P/Q, Q monic, by interpolation and the Euclidean remainder sequence, checked on strengths held out: this gives the common denominator Q_c of the entries and the degree of the numerators over it, $B = \tilde{P}/Q_c$. On

every further prime, Q_c and all the $\tilde{P}$ at once from $\deg \tilde{P}+6$ strengths: the values of each entry times Q_c lie on a polynomial of degree at most $\deg \tilde{P}$, which is one linear condition on Q_c from each entry; then each $\tilde{P}$ by interpolation, checked at every strength. Over $\mathbb{Q}$, every coefficient by Chinese remaindering and rational reconstruction with one common denominator, taken when two rounds agree.

(c) *Acceptance.* B agrees with an elimination of the whole box, not the reduced one, at a prime and a strength drawn at random.

(4) *The law over* $\mathbb{Z}$. Modulo primes, with no elimination: B over its common denominator at consecutive points x, the rows carried along by $w_i(x) = w_{i-1}(x-1)\,B(x)$, the dependence of $w_0, \dots, w_h$ at each point (Theorem 23.5(iii)), each ratio d_i/d_h reconstructed as a rational function, denominators cleared; over $\mathbb{Q}$ by Chinese remaindering until two rounds agree; integral and primitive; checked at a prime and a point drawn at random.

(5) *The value.* Theorem 21.1 on the law: the native gauge $\alpha^\star$ on the orthant and $\det H$ by Newton's method, the density operator and its germ at $f^\star$, with the exponent $\frac{n-3}{2}$ checked against the law, and the constant $g(0)$ of Theorem 21.1(i); the Beta sums $\langle 1\rangle_K = \sum_m e_m \int_0^{f^\star} t^K (f^\star - t)^{\frac{n-3}{2}+m} dt$ at h consecutive large strengths K; the law run back to $k=1$. In ball arithmetic, which carries every rounding, from two starting strengths, the precision doubled until the two readings agree to the digits wanted and their balls are that narrow.

(6) *Checks, none of them part of the construction.* The offset closings $X_j(b) = [\alpha^0]\, bL^j$ obey $X_j(b_i) = \sum_l B_{il}(1-j)X_{j-1}(b_l)$, the one-step law on the torus at $k=-j$, exactly; the closings obey the reflected law (Corollary 18.2); the spectrum of $B(\infty)$ against the native values found by other means (Proposition 23.11), and where B is not finite at infinity the limits of its eigenvalues against a census of native gauges completed by continuation in the exponents from generic ones (Proposition 23.18(iv)); the law against one read off closings, where there is one; B and the law again at primes and points drawn afresh.

Remark 23.13 (Rays with an origin; directions with rational masses). [derived] Everything above holds for a ray with an origin $(u_0; a_0, b_0)$ of degree zero, $|u_0| = a_0 r + b_0(n-r)$, whose values are $\langle g\rangle_k = \int_{y_{e_0}=0} g\, G_0\, f^k\, dy$ with $G_0 = \alpha^{u_0} K^{-a_0} \Psi^{-b_0}$ (Def. 35.1 with the origin form $G_0\, dy$). The mark relations gain the defect of the origin, $c_e^{(0)} = u_{0,e} - a_0 R_e - b_0 R_e^*$: in Proposition 23.3(ii) the factors $km_{1,e} + u_e$, $ks + a$, $kt + b$ become $km_{1,e} + u_e + u_{0,e}$, $ks + a + a_0$, $kt + b + b_0$, with the same proof at the data $(k\mathbf{m}_1 + u + u_0;\ ks + a + a_0,\ kt + b + b_0)$, and nothing else changes; Thm. 35.19 is stated for rays with an origin, so Theorem 23.10 holds, with $A \geq 1$ when $a_0 \neq 0$ and $B \geq 1$ when $b_0 \neq 0$. In Theorem 21.1 only the constant of (i) changes: the density is $\varrho_0(t) = \int_{f=t} G_0\, d\sigma/|\nabla f|$, G_0 is analytic and positive on the chart, the Morse lemma integrates $G_0 J$ in place of J, whose odd moments over a ball vanish as well, and $g(0)$ is multiplied by $G_0(\alpha^\star)$; (ii)–(v) use only that ϱ_0 is a density of the values, carried by $(0, f^\star]$ and analytic inside. So a direction $(\mathbf{m}; s, t)$ with rational masses and integral s, t, strictly inside its walls, is read as an integral direction $(\mathbf{m}'; s, t)$ next to it and strictly inside the walls, with the origin $(\mathbf{m} - \mathbf{m}'; 0, 0)$, at $k=1$, where $\langle 1\rangle_1 = P_{\mathcal{B}}(\mathbf{m}; s, t)$. Scaling the direction by a common denominator N and reading at $k = \frac{1}{N}$ gives the same number, the law holding at every k (Corollary 23.6); but the scaled direction has the stride f^N, and its box of wall degree T is the box of wall degree NT of f, while the origin keeps the stride of the integral neighbour.

The program. `rayreader` [computed] carries out steps (1)–(5) from the list and the direction alone:

```
python3 rayreader.py V "1 1 1 1 1 1 1 1 ; 0 2" --digits 30
```

Each step, and within steps (3) and (4) each prime, is saved in a run directory as it finishes, so a run stopped at any point resumes with the same command, and the primes and strengths are spread over

the processors. The eliminator is a C program that reduces the rows in the order of their leading column, holds sums of products in 128 bits and reduces them only when a column is read, and visits the touched columns of a row through a bitmap. On V at its lean the targets need 19581 of the 87507 relations, and an elimination of those takes 4.5 s on one core. Table 23.14 lists what it returns on every ray whose value is known by other means, and on V.

What is certain. The relations have integer coefficients and every computation modulo a prime is exact. Chance enters through the strengths and primes drawn. A strength modulo q gives the generic answer, the answer over $\mathbb{Q}(k)$ reduced, unless it is a zero of one of finitely many nonzero polynomials fixed by the box (pivots of the generic elimination, denominators of B); so the masters, the relations kept and the values of B are the generic ones off a set of strengths of density at most the total degree of those polynomials over q. A reconstruction of B that is wrong agrees with the whole box at the drawn prime and strength only if the strength is a zero of the reduction of a nonzero polynomial over $\mathbb{Z}$, the numerator of the difference, or a degenerate strength, or the prime divides every coefficient of that numerator; the same holds for the law. The value is computed in ball arithmetic, so every rounding is accounted for; the truncations of the germ and of the Beta sums are not bounded by a proof, but by the size of their last terms, and the error given is the difference of the readings from two starting strengths, an estimate.

What it costs. The box has $O(T^{n-1})$ insertions per level (Lemma 23.2(iv)), divided by up to the number of symmetries; a seed's mark relation has $1 + |\mathcal{B}|$ terms. Step (3) is one elimination per strength; the first prime needs about $\deg P + \deg Q$ strengths, the others $\deg \tilde{P} + 6$, and the number of primes is about the height of the coefficients of $\tilde{P}$ and Q_c, numerators and common denominator together, over 62 bits. Step (4) needs no elimination: per prime, about $2D + 2h$ points, each a product of an $h \times \mu$ and a $\mu \times \mu$ matrix, for a law of order h and degree D, and the number of primes is about twice the height of the law over 62 bits. Step (5) works at a precision of about 3.3 bits per digit wanted, plus the digits lost to the growth of the other solutions of the law when it is run back.

ray	value read	digits	against	law (h, D)	μ	time
$\binom{[3]}{1}$, $(\mathbf{1}; 1, 1)$	2.34390723868945889…	67	ℓ (§19)	$(2, 2)$	2	< 1 s
$\binom{[4]}{1}$, $(\mathbf{1}; 1, 1)$	8.41439832211715999…	65	$7(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$	$(2, 3)$	2	< 1 s
$\mathrm{M}(K_4)$, $(\mathbf{1}; 0, 2)$	7.21234141895756571…	63	$6(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, Thm. 20.15	$(2, 5)$	2	< 1 s
$\binom{[5]}{1}$, $(\mathbf{1}; 1, 1)$	39.9455880119962456…	59	§22, share side	$(3, 4)$	3	< 1 s
H, $(\mathbf{1}; 1, 1)$	4.09623699833788285…	66	§22, gauge end	$(4, 7)$	4	< 1 s
H, $((1, 1, 3, 3); 3, 1)$	0.48830319489526888…	61	Ex. 20.12, §22	$(4, 25)$	4	2 s
N, $(\mathbf{1}; 0, 2)$	4.75680256375822944…	63	§22, gauge end	$(6, 24)$	7	1 s
N, $(\mathbf{1}; 1, 1)$	4.57891247888240480…	63	Prop. 20.36(vi)	$(11, 52)$	15	5 s
$\binom{[3]}{1}$, $(\frac{1}{2}, 1, \frac{3}{2}; 1, 1)$	4.77739479233093029…	65	quadrature	$(3, 4)$	4	< 1 s
V, $(\mathbf{1}; 0, 2)$	7.12823937562968675…		tropical 7.124 ± 0.009	$(24, 377)$	60	45 min

Table 23.14. What the program returns from the list and the direction, at 30 digits asked, on two processors.

Reading the table [computed]. "Digits" is $-\log_{10}$ of the distance to the value it is set against, which the program does not see; the program computes beyond the digits asked, and its own estimate of its error, the difference of two starting strengths, is below 10^{-47} on every row. For V no value is known to more than three digits; the tropical estimate of Proposition 20.37(v) is a sampling, and the sixty digits of Proposition 23.18(vi) are a second reading of the same law. The values set against are closed forms for the first three rows; for $\binom{[5]}{1}$ the share side of §22; for H and N the gauge end of §22 run on laws found there from closings, which the program does not use, its law being constructed from the relations. The row with rational masses is read through the origin $(-\frac{1}{2}, 0, \frac{1}{2}; 0, 0)$ on the centre ray (Remark 23.13) and set against a quadrature: on the chart $\alpha_3 = 1$, with $\alpha_1 = u^2$, the

integral over α_2 is elementary, and $P_{\binom{[3]}{1}}(\frac12,1,\frac32;1,1)=\int_0^\infty 2\log\big((1+u^2)^2/u^2\big)\,\frac{du}{1+u^2+u^4}$. The time is for all five steps.

23.6 Receipts

Example 23.15 (The three-mark centre, without closings)**.** [derived]; the elimination [computed] exactly. Take $\binom{[3]}{1}$ at $(\mathbf{1};1,1)$, so $L=e_1e_2/e_3$, all walls are 1 and the wall degree of α^u is the sum of its positive exponents. The box $\mathfrak{B}_2$ has 30 insertions modulo the six permutations and Cremona, at the four levels, and $\mathcal{R}_2$ has 32 relations. One elimination leaves two masters,

$$b_0=1,\qquad b_1=\frac16\sum_{i\neq j}\frac{\alpha_i}{\alpha_j},$$

and both targets reduce:

$$B(k)=\begin{pmatrix}3 & 6\\ \dfrac{2(k-1)}{k-2} & \dfrac{(k-1)(7k-11)}{(k-2)^2}\end{pmatrix},\qquad V(k-1)=B(k)V(k),\quad V=\big(P(k),\langle b_1\rangle_k\big).$$

The first row needs no relation: $L=3+\sum_{i\neq j}\alpha_i/\alpha_j=3+6b_1$ is an identity of Laurent polynomials. The second has the certificate of Theorem 23.5(ii) with G_2,G_3 the images of G_1 under the transpositions $(1\,2)$, $(1\,3)$ and

$$G_1=\frac{\alpha_2^2+\alpha_3^2}{6(k-2)\,\alpha_1^2}+\frac{(5k^2-6k-4)\,\alpha_2\alpha_3}{6(k+1)(k-2)^2\,\alpha_1^2}+\frac{3k-4}{6(k+1)(k-2)^2}\Big(\frac{\alpha_3^2}{\alpha_1\alpha_2}+\frac{\alpha_2^2}{\alpha_1\alpha_3}\Big)$$
$$+\frac{5k-9}{6(k-2)^2}\,\frac{\alpha_2+\alpha_3}{\alpha_1}+\frac{3k-4}{6(k+1)(k-2)}\Big(\frac{\alpha_3^2}{\alpha_2^2}+\frac{\alpha_2^2}{\alpha_3^2}\Big),$$

a Laurent polynomial supported on five orbits of seeds, found by one linear solve on the seeds and verified as an identity of rational functions of (α,k); at $\alpha=(1,2,4)$, $k=\frac13$ both sides equal $\frac{11903}{800}$. The cyclic vector: $P(k-1)=3P(k)+6\langle b_1\rangle_k$, and the second row,

$$P(k-2)=3P(k-1)+6\Big(\frac{2(k-1)}{k-2}P(k)+\frac{(k-1)(7k-11)}{(k-2)^2}\langle b_1\rangle_k\Big),$$

with $6\langle b_1\rangle_k=P(k-1)-3P(k)$ eliminated and $(k-2)^2$ cleared, is

$$(k-2)^2P(k-2)-(10k^2-30k+23)\,P(k-1)+9(k-1)^2P(k)=0,$$

since $3(k-2)^2+(k-1)(7k-11)=10k^2-30k+23$ and $12(k-1)(k-2)-3(k-1)(7k-11)=-9(k-1)^2$. At $k\mapsto k+2$ this is (L) of §19, with $10(k+2)^2-30(k+2)+23=10k^2+10k+3$. So (L) holds identically in k (Corollary 23.6), which is §19, Steps 3–5, in one elimination. At infinite strength

$$B(\infty)=\begin{pmatrix}3&6\\2&7\end{pmatrix},\qquad B(\infty)\begin{pmatrix}1\\1\end{pmatrix}=9\begin{pmatrix}1\\1\end{pmatrix},\qquad B(\infty)\begin{pmatrix}1\\-\frac13\end{pmatrix}=\begin{pmatrix}1\\-\frac13\end{pmatrix},$$

and $(1,1)$ and $(1,-\frac13)$ are the values of (b_0,b_1) at the table's gauge $(1,1,1)$ and at the sign $(-1,1,1)$, of native values $\frac19$ and 1 (§19, Step 6). They span, so by Proposition 23.11(ii) the two masters are independent: the symmetric part of the ray has dimension exactly two, and no law of order one follows from the relations.

ray	box	relations	unknowns	μ	d_B	law (h, D)	time	checks
$\binom{[3]}{1}$, $(\mathbf{1}; 1, 1)$	$(2; 1, 1)$	32	30	2	2	$(2, 2)$	0.2 s	(L) of §19; $x_{\le 12}$; $\{9, 1\}$
$\binom{[4]}{1}$, $(\mathbf{1}; 1, 1)$	$(2; 1, 1)$	46	42	2	3	$(2, 3)$	0.2 s	$x_{\le 25}$; $\{16, 4\}$
$\binom{[5]}{1}$, $(\mathbf{1}; 1, 1)$	$(3; 1, 1)$	112	100	3	5	$(3, 4)$	0.3 s	$x_{\le 25}$
$\binom{[3]}{1}$, $(2\mathbf{1}; 4, 1)$	$(2; 1, 1)$	124	79	2	4	$(2, 6)$	0.1 s	§18.3; $x_{\le 14}$; $\{243, -\frac{64}{27}\}$
H, $(\mathbf{1}; 1, 1)$	$(2; 1, 1)$	180	102	4	3	$(4, 7)$	0.6 s	§18.4; $x_{\le 14}$; $\{25, 9, 1, 1\}$
$\mathrm{M}(K_4)$, $(\mathbf{1}; 0, 2)$	$(2; 0, 1)$	462	211	2	8	$(2, 5)$	0.6 s	Thm. 20.15; $x_{\le 12}$; $\{256, 4\}$
H, $((1, 1, 3, 3); 3, 1)$	$(2; 1, 1)$	1000	631	4	9	$(4, 25)$	3.3 s	Ex. 20.12; $x_{\le 14}$
$\binom{[5]}{2}$, $(\mathbf{1}; 1, 1)$	$(3; 1, 1)$	245	234	5	8	$(4, 6)$	0.5 s	$x_{\le 20}$; $\{100, 4, 4, 1, -8\}$
N, $(\mathbf{1}; 0, 2)$	$(2; 0, 1)$	728	329	7	6	$(6, 24)$	2.5 s	Thm. 20.35; $x_{\le 16}$; see below
N, $(\mathbf{1}; 1, 1)$	$(3; 1, 1)$	3774	1909	15	17	$(11, 52)$	24 s	Prop. 20.36(ii); $x_{\le 24}$; see below
V, $(\mathbf{1}; 0, 2)$	$(3; 0, 1)$	87507	32810	60	83	$(24, 377)$	$50\,\mathrm{s} + 4.5\,\mathrm{s}/k$	Proposition 23.18
V, $(\mathbf{1}; 1, 1)$	$(3; 1, 1)$	98622	54740	119			$563\,\mathrm{s} + 480\,\mathrm{s}/k$	Proposition 23.19

Table 23.16. The construction on every ray met.

Reading the table [computed]. The box is $(T; A, B)$, with A, B the units of the two letters; relations and unknowns are counted after the symmetries; μ is the number of masters and d_B the largest degree in k of an entry of $B(k)$, numerator or denominator; (h, D) are the order and degree of the law of Theorem 23.5(iii); the time is for the whole construction, system, eliminations at 40 to 60 strengths modulo $p = 2^{61} - 1$, reconstruction and law, on one core, and for V the system and masters and then each strength (Table 23.14 has the whole reading). Checks: a section or theorem named means that the law equals the law printed there, as polynomials modulo p up to a common factor (for N at the lean, the law of the receipts of §20, of integer coefficients; at its centre, the law of order 11 of Proposition 20.36(ii), which was only known modulo two primes, through its symbol); $x_{\le J}$ means that the closings $x_0, \dots, x_J$, computed independently by truncated expansion, satisfy the reflected law (Corollary 18.2); a set is the spectrum of $B(\infty)$, exact, equal to the reciprocals of the roots of the symbol together with the eigenvalues of the masters that the cyclic vector of 1 does not reach (one more 4 for $\binom{[5]}{2}$). For $\mathrm{M}(K_4)$ the spectrum is that of the conjugated matrix, and the limit of the spectrum of $B(k)$ itself (§23.4). For $\binom{[5]}{1}$ the matrix is not finite at infinity, and Proposition 23.11(i) says why: its masters are 1, $b_1 = \frac{1}{20}\sum_{i \ne j} \alpha_i/\alpha_j$ and $b_2 = \frac{1}{20}\sum_{i \ne j} \alpha_i^2/\alpha_j^2$, and $b_2 = 1$ at every sign, so the masters' values at the three classes of signs, of native values $\frac{1}{25}, \frac{1}{9}, 1$, lie in a plane, where three eigenvectors of distinct eigenvalues cannot. For N,

$$\det\big(\lambda - B(\infty)\big) = (\lambda - 24)^2(\lambda^2 - 324\lambda - 10800)(\lambda^3 + 28\lambda^2 - 320\lambda + 1728) \qquad (\mathbf{1}; 0, 2),$$
$$\det\big(\lambda - B(\infty)\big) = (\lambda - 361)(\lambda - 25)^2(\lambda - 9)(\lambda - 1)^8(\lambda + 80)\big(\lambda^2 + \tfrac{34}{3}\lambda - \tfrac{361}{3}\big) \qquad (\mathbf{1}; 1, 1),$$

the reciprocals of the roots of the symbols of Theorem 20.35(ii) and Proposition 20.36(ii), with one more 24 at the lean and, at the centre, one more 25 and three more 1.

Exact certificates [computed]. For the three-mark centre and lean, the bananas on four and five marks, H at its centre and on the ray of Example 20.12, the fixture, $\binom{[5]}{2}$ at its centre and N at its lean the certificate of Theorem 23.5(ii), taken symmetric under the automorphisms of the list that fix the masses, is found by one linear solve on the seeds (reduced echelon form, free unknowns 0), modulo one prime or several, reconstructed over $\mathbb{Q}$, and verified as an identity of rational functions of (α, k): the identity times $K\Psi$ and a common denominator, monomial by monomial, each coefficient a polynomial in k over $\mathbb{Q}$. The numbers of nonzero coefficients are 5, 40, 8, 37, 145, 535, 261, 234 and 1016, in that order, over 1 to 3 primes. The one-step matrices so certified give the laws exactly:

from them, the law (L) of §19, the laws of §18.3 and §18.4, of Theorem 20.15, of Example 20.12 and of Theorem 20.35 follow as identities of polynomial vectors over $\mathbb{Q}$ (Theorem 23.5(iii)). So those laws are proved at every k (Corollary 23.6), with certificates that nothing was guessed to find.

Example 23.17 (The fixture, bracketed by the method of §19)**.** [derived]; the table [computed] exactly. At $(\mathbf{1};0,2)$ on $\mathrm{M}(K_4)$ the constructed law is that of Theorem 20.15,

$$-k^5P(k)+2(2k+1)(65k^4+130k^3+105k^2+40k+6)\,P(k+1)\\ -16(k+1)(2k+1)(2k+3)(4k+3)(4k+5)\,P(k+2)=0,$$

now with an exact certificate, so it holds at every k (Corollary 23.6). Run §19 on it. The corner (Corollary 18.2, with Cor = 84 from Theorem 20.15(i) and $[k^5]c_0=-1$, $c_1(0)=12$, $c_2(0)=-720$) gives $P(2)=\frac{P(1)-7}{60}$. With $P(K)=c(K)P(1)+r(K)$, $c(1)=1$, $c(2)=\frac{1}{60}$, $r(1)=0$, $r(2)=-\frac{7}{60}$, positivity gives $P(1)>a_K=-r(K)/c(K)$ wherever $c(K)>0$, and the gauge, $0<f\le f^\star=\frac{1}{256}$ with equality at one point (Thm. 35.10(i)), gives $P(K)<\frac{1}{256}P(K-1)$, so $P(1)<b_K=\frac{r(K-1)-256\,r(K)}{256\,c(K)-c(K-1)}$ wherever the denominator is positive. In exact arithmetic both signs hold for $2\le K\le 15$, the width falls by about 64 per step, the ratio of the native values $\frac{1}{4}$ and $\frac{1}{256}$, and at $K=12$

$$\frac{12531162726888825592963 9}{17374611099178960704000}<P_{\mathrm{M}(K_4)}(\mathbf{1};0,2)<\frac{29090775086028604309225 7947}{40334717113590600816192000},$$

a bracket of width $3.50\cdot10^{-19}$. As a check outside the bracket, six times the Trailmoment of the counting input at index three, $6(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))=7.2123414189575657124\ldots$ [recalled], lies in it. So the examination of §19 runs on the fixture with nothing guessed: law and certificate by Theorem 23.5, corner by Corollary 18.2, and the two inequalities.

Proposition 23.18 (V at its full lean)**.** [computed], *modulo* $p=2^{61}-1$ *unless said otherwise. Let* V *be the eight-mark list of Definition 20.33 at the direction* $(\mathbf{1};0,2)$, $L=\Psi_V^2/\alpha^{\mathbf{1}}$.

(i) The box. $\mathfrak{B}_3$, *one unit of* Ψ, *modulo the* 64 *automorphisms of* V*:* 87507 *relations on* 32810 *insertions,* 9723 *of them Laurent seeds. One elimination leaves* 60 *masters, the Laurent insertions of wall degree at most* 2 *left free, and every target reduces to them; the system and the masters take* 50 *s. The targets need* 19581 *of the relations (Procedure 23.12(3)(a)), and* $B(k)$ *at each further strength takes* 4.5 *s on one core.*

(ii) The torus. *The offset closings* $X_j(b)=[\alpha^{j\mathbf{1}-u}]\,\Psi_V^{2j}$ *of the* 60 *masters* $b=\alpha^u$, $j\le5$, *computed exactly by a truncated expansion, with the closings* $x_j=1$, 64, 34080, 39638080, 65433978400, 131819906744064 *of Proposition 20.37(iv) among them, satisfy*

$$X_j(b_i)=\sum_l B_{il}(1-j)\,X_{j-1}(b_l)\qquad(0\le i<60,\ j=2,3,4,5),$$

the one-step law at the strengths $k=-1,-2,-3,-4$ *on the torus, with* B *computed at those strengths directly:* 240 *identities, none trivial.*

(iii) The law. *At* 62 *consecutive strengths, the rows* $w_0,w_1,\ldots$ *of Theorem 23.5(iii) have ranks* $1,2,\ldots,24$ *and then* 24 *through* w_{61}*, and the same from a base point* 20 *strengths lower: the law of* $\langle1\rangle_k$ *has order* 24*. The cyclic space of* 1 *is* 24 *of the* 60 *dimensions of the masters. From* B *over* $\mathbb{Q}(k)$*, (v), the law over* $\mathbb{Z}$ *(Procedure 23.12(4)) has degree* 377 *and coefficients of up to* 753 *digits, from* 63 *primes, and it holds at a prime and a point drawn at random.*

(iv) The spectrum. $B(k)$ *is not finite at infinity. Over* $\mathbb{Q}$ *at* $k=10^5$, $2\cdot10^5$, $3\cdot10^5$ *(Chinese remaindering over* 56 *primes, stable from* 52*; numerators of up to* 1571 *bits, denominators*

of up to 1513*) its largest entries are* $1.3 \cdot 10^{16}$, $1.1 \cdot 10^{17}$, $4.0 \cdot 10^{17}$, *growing like* k^3, *and the eigenvector of the eigenvalue next to* $1/f^\star$, *normalized at the master* 1, *lies at relative distance* 675, 1440, 2220 *from the masters' values at* $\alpha^\star$. *So Proposition 23.11(i) does not apply, and no change of masters as in (iii) is known. The eigenvalues settle all the same: eigenvalues that close in on one another are taken together, and the mean of each group, a symmetric function and so analytic in* $1/k$, *is extrapolated to infinite strength from the three strengths. Against the limits stand the critical points of* f. *Newton's method on* $\log f$, *from about* 10^5 *random complex starts and* $1.6 \cdot 10^5$ *starts on the fixed loci of the abelian groups of automorphisms twisted by characters, finds* 327 *gauges, all nondegenerate, in* 30 *orbits of the automorphisms, with* 18 *native values; and a continuation in the exponents finds the same: the product* $\alpha^\kappa \Psi_V^{-\lambda}$ *with generic exponents,* κ *constant on* $P_1 \cup P_2$ *and on* $P_3 \cup P_4$ *so that it is invariant under the automorphisms, has* 463 *critical points, and continued to the exponents of the stride,* 327 *of their paths end at the* 327 *gauges, one each, and the other* 136 *run off, their coordinates reaching moduli between* e^5 *and* e^{11} *(against the eighth) where they were left, and at most* $e^{2.2}$ *at the gauges. Each native value, counted once for each of its orbits, claims a group of the limits,* 30 *of the* 60, *every group's limit agreeing with the mean of its reciprocal native values to less than the difference of the two- and three-point extrapolations (*$1/f^\star$ *to* $2.4 \cdot 10^{-9}$ *relative,* $1/0.000864231320476$ *to* $8 \cdot 10^{-11}$, $1/0.416392354$ *to* $3 \cdot 10^{-11}$*):*

one orbit: $f^\star$, 0.000818456, 0.000864231, 0.124846403,
0.870337682, $-0.592832429 \pm 0.713227829\,i$,

two orbits: 0.004487298, -0.013533898, $0.007758455 \pm 0.013607932\,i$, -0.026238829,
0.066968510, 0.109146811, $\frac{1}{8}$, 0.416392354, 0.870512702,

three orbits: 0.021107646.

Eleven of the 18 *values are the census of Proposition 20.37(iii). Of the seven new ones,* 0.870512702 *lies next to the conjugate gauge: the three eigenvalues near* 1.1488, *one tending to* $1/0.870512702$ *like* $1/k$ *and two approaching like* $k^{-1/2}$, *have the mean of* $1/0.870337682$ *and twice* $1/0.870512702$ *to* $1.6 \cdot 10^{-7}$. *The other* 30 *limits have the reciprocals*

$\frac{1}{4}$, $\frac{3}{16}\,(2)$, $\frac{1}{16}\,(3)$, $-\frac{1}{16}\,(2)$, $\frac{3}{64}\,(2)$, $\frac{1}{48}\,(6)$, $-\frac{1}{48}$, $\frac{1}{64}\,(4)$,
$-\frac{1}{64}\,(2)$, $-\frac{1}{80}\,(2)$, 0.97307, 0.41639, 0.19479, -0.020346, 0.0052756,

the rational ones to six to nine digits, and 0.41639 *a third limit at the value of the two orbits of* 0.416392354, *to the accuracy of its extrapolation. Along the* 136 *paths that run off,* f *is* $\frac{3}{64}$, $\frac{1}{16}$, $\frac{1}{64}$ *or* $-\frac{1}{64}$ *to* 10^{-5} *where they were left, at* $1 - t \le 1.3 \cdot 10^{-4}$*: four of these values. So the direction lies in the exceptional set of Thm. 35.19(i), where native gauges are lost (with exponents generic in every mark the product has at least* 751 *critical points, against the* 327 *of* f*). The* 30 *are not claimed as native values; four of their values are those of* f *along the critical points that the direction loses. On* N *at its lean, where* B *is finite at infinity, the same census finds* 35 *gauges in* 7 *orbits, as many as the product with generic exponents has critical points, and the* 7 *eigenvalues of* $B(\infty)$ *are their reciprocal native values,* 24 *twice for the two orbits of* $\frac{1}{24}$. *The largest limit is* $1/f^\star$, *simple, and the next has modulus* 1221.8*: if a change of masters as in Proposition 23.11(iii) exists, every nondegenerate native value other than* $f^\star$ *has modulus above* $f^\star$, *which is what positivity asks of the ray.*

(v) The matrix over $\mathbb{Q}(k)$ *(Procedure 23.12(3)). Modulo a prime,* 178 *random strengths determine every entry:* 72 *entries vanish, and in lowest terms the numerators have degree at most* 83 *and the denominators at most* 80. *All the denominators divide one polynomial of degree* 85,

$$Q_c(k) = k\,(k-1)^2\big(k-\tfrac{3}{2}\big)(k-2)^8\big(k-\tfrac{5}{2}\big)(k-3)^7\;S(k),$$

over which the numerators have degree at most 88*, so that each further prime needs* 94 *strengths. Over* $\mathbb{Q}$*, from* 9 *primes, the coefficients of the numerators over* Q_c *and of* Q_c*, monic, have up to* 274 *bits; the factor* S *is irreducible over* $\mathbb{Q}$*, of degree* 65*, with integer coefficients of up to* 226 *bits in its primitive form.* B *agrees with an elimination of the whole box at a prime and a strength drawn at random.*

(vi) The value *(Procedure 23.12(5)). The native gauge on the orthant has* $f^\star = 2.37093045489\ldots \cdot 10^{-4}$ *(Proposition 20.37(ii)) and* $\det H = 2.40551463730\ldots \cdot 10^{-3}$*; the law's density operator has order* 377*, its symbol vanishes at* $f^\star$*, and the exponent there read from the law is* $\frac{5}{2} = \frac{n-3}{2}$*. With* 60 *digits asked, from the starting strengths* 93 *and* 118*, at* 3176 *bits,*

$$P_V(\mathbf{1}; 0, 2) = \langle 1 \rangle_1 = 7.12823937562968675472229135353893457156078898940720766490704\ldots,$$

the two readings differing by $2.4 \cdot 10^{-82}$*; with* 30 *digits asked, from* 57 *and* 82*, the same to* 34 *digits. So* $P(1)/\mathrm{Cor} = 0.0237607979187656\ldots$*, and the tropical* 7.124 ± 0.009 *of Proposition 20.37(v) is* 0.0042 *off. The corner, which the program does not use, checks the law and the values together:* c_0 *vanishes to order* $7 = n - 1$ *at* $k = 0$*, and with* $\mathrm{Cor} = 300$ *(Proposition 20.37(i)) the relation of Corollary 18.2,*

$$[k^7]c_0 \cdot 300 + \sum_{i=1}^{24} c_i(0)\, P(i) = 0,$$

holds to 73 *digits: with* 40 *digits asked the left side is* $1.5 \cdot 10^{623}$*, against terms of up to* $2.4 \cdot 10^{696}$*. From the list and the direction to this value the program takes* 45 *minutes on two processors, from scratch:* 2 *for the masters,* 39 *for* B *over* $\mathbb{Q}(k)$*,* 3 *for the law and seconds for the value.*

So V *at its lean has a law, constructed by Theorem 23.5, where Proposition 20.37(vi) had none, and it is read: the closings that it would have taken some hundreds of to guess are replaced by one elimination per strength. Its matrix over* $\mathbb{Q}(k)$ *is reconstructed modulo primes and checked at random points; its certificate, which Theorem 23.5(ii) provides, is not computed (Ledger, O1).*

Proposition 23.19 (V at its centre)**.** [computed] *modulo* $p = 2^{61} - 1$*. At* $(\mathbf{1}; 1, 1)$*,* $L = K_V \Psi_V / \alpha^{\mathbf{1}}$*, the box* $\mathfrak{B}_3$ *with one unit of each letter, modulo the* 64 *automorphisms and Cremona, has* 98622 *relations on* 54740 *insertions, with* 5163 *Laurent seeds. One elimination leaves* 119 *masters, the Laurent insertions of wall degree at most* 2 *left free, and every target reduces to them at three random strengths:* V *at its centre has a one-step law of size* 119*. The offset closings* $X_j(b) = [\alpha^{j\mathbf{1}-u}]\,(K_V \Psi_V)^j$ *of the* 119 *masters, computed exactly for* $j \le 5$ *(the closings themselves are* $x_j = 1$*,* 65*,* 34337*,* 39933281*,* 65997650593*,* 133148608004065*), satisfy the one-step law on the torus at* $k = -1$ *and* $k = -2$*, with* B *computed at those strengths directly:* 238 *identities, none trivial. The system and the masters take* 9.4 *minutes, and* $B(k)$ *at each further strength about* 8 *minutes with the eliminator of the receipts, on the whole box; the relations the targets need, the matrix over* $\mathbb{Q}(k)$*, its order and its spectrum are* [not yet computed].

23.7 What it shows

(i) *No step of the law is a guess.* The masters are what one elimination leaves, the matrix is what it returns, and the law is the first dependence of a cyclic vector. Every step is an identity among values of insertions, each relation being Thm. 8.3 at the data of the ray, so the law holds identically in k, the corner relation and the closings' recurrence follow from it (Corollary 18.2), and its certificate is the record of the elimination, supported on the seeds. The two halves of step (3) of Procedure 18.6,

the recurrence of the closings and the ansatz for a certificate, become one computation with nothing to choose but the box, and the box is checked.

(ii) *The structure is the list's, the size is the walls'.* What the closings could not see is that the ray is a module of finite dimension presented by the relations of the period, and that its presentation lives in one unit of each letter above the Laurent insertions (§23.2). The masters are set by the list and the direction; the box that reaches them is set by the walls, through the wall degree, and in every case met it is the same box: one unit of each letter and $T \leq 3$, from three marks to eight. The symmetries of the list divide the work by their number.

(iii) *Carry the matrix.* The scalar law is a projection of the matrix, and its size is the projection's, not the structure's. At N's centre the entries of B have degree at most 17, and the law has order 11, degree 52 and coefficients of 88 digits, where the law of order 12 read off closings has coefficients of 1300 digits (Proposition 20.36(v)). At V's lean the entries of B have degree at most 83 over denominators of degree at most 80, and the law has order 24 and degree 377. The matrix, the certificate and the spectrum are what the construction delivers at the size of the structure, and the matrix is the law.

(iv) *For §19 in general.* §19 needed four inputs: the law, its certificate and its validity down to $k \to 0^+$, the corner, and positivity. The first three are now one construction (Theorem 23.5, Corollary 23.6), with the native values from the matrix where it is finite at infinity or made so (Proposition 23.11). The corner is Corollary 18.2 applied to the constructed law, and positivity is Theorem 18.5, both unchanged. What remains per ray is what was always per ray: the signs of the rational solutions when the law has more than one master (Ledger L.2, O2), or the gauge end (Theorem 21.1), which reads any ray whose law is known and needs neither corner nor positivity.

(v) *V, and what sets the cost.* The list that §20 could not read is read from its list and its direction: the box $(3; 0, 1)$ and 60 masters, B over $\mathbb{Q}(k)$, the law of order 24 and degree 377 over $\mathbb{Z}$, and $P_V(\mathbf{1}; 0, 2) = 7.12823937562968675\ldots$ by the gauge end (Proposition 23.18(v)–(vi)), in 45 minutes on two processors. The cost is set by B: the degree of its entries in k fixes the strengths per prime, and the height of its coefficients the number of primes, each strength one elimination of the relations the targets need. Nothing in the procedure depends on the closings, on a guess, or on the number of marks except through these. Two things are not read off the matrix. Its spectrum at infinite strength, at V's lean, is reached only through limits (Proposition 23.18(iv)), half of them claimed by native values and half not, the direction being exceptional; a change of masters as in Proposition 23.11(iii) is not found, and with it the hypothesis of positivity, which the gauge end does not need. And the error of the value is an estimate, the difference of two starting strengths (Ledger, O23).

L Ledger, updated by this entry

L.2, Open. The items changed by this entry, with what stands of them.

- **O1** *(Certificates.)* Every law of Table 23.16 is constructed with its certificate (Theorem 23.5(ii)) and holds as an identity in k, so at every real $k > 0$ and in Corollary 18.2 (Corollary 23.6). Certified exactly, with the certificate verified over $\mathbb{Q}$ and the law derived from the certified matrix (§23.6): the law (L) of §19, the laws of §18.3 and §18.4, of Theorem 20.15, of Example 20.12 and of Theorem 20.35. The bounds on the pieces down to $k \to 0^+$ are not needed: the identity in k comes from Thm. 8.3 through Proposition 23.3, and the corner is Corollary 18.2 applied to the law. The converse of Corollary 18.2 is no longer used: laws are constructed, not reflected. Open: exact certificates for N's centre, whose certificate coefficients need more than 80 strengths per prime, and for V at its lean, whose matrix over $\mathbb{Q}(k)$ is reconstructed and checked at random

points (Proposition 23.18(v)) but whose certificate is not computed; and the banana for every n by an operator proof. [not yet computed]

- **O10** *(N and V.)* V at $(\mathbf{1}; 0, 2)$: 60 masters, every target reduces in $\mathfrak{B}_3$, the one-step law holds on the torus at $k = -1, \dots, -4$, and the law of $\langle 1 \rangle$ has order 24. $B(k)$ is not finite at infinity; 30 of the limits of its eigenvalues are the reciprocals of the 18 native values, each once for each of its orbits, seven of them new to the census of Proposition 20.37(iii), the census now complete as far as a continuation in the exponents reaches; the other 30 no native gauge accounts for, the direction being exceptional in the sense of Thm. 35.19(i), and four of their values are those of f along the critical points that the direction loses (Proposition 23.18(iv)). V at $(\mathbf{1}; 1, 1)$: 119 masters, every target reduces in $\mathfrak{B}_3$, and the one-step law holds on the torus at $k = -1, -2$ (Proposition 23.19). The law of order 11 of N's centre is constructed, with its symbol confirmed (Table 23.16); its exact coefficients by Chinese remaindering stay [not yet computed]. V at $(\mathbf{1}; 0, 2)$ is read: B over $\mathbb{Q}(k)$, the law of order 24 and degree 377 over $\mathbb{Z}$, and $P(1) = 7.12823937562968675\dots$ by the gauge end, against the tropical 7.124 ± 0.009 of Proposition 20.37(v) (Proposition 23.18(v)–(vi)). Open: a change of masters for V at its lean as in Proposition 23.11(iii), which would put every nondegenerate native value among the limits; what the other 30 limits are; B over $\mathbb{Q}(k)$ and the value at V's centre; the rest of the item as it stood.

- **O21** *(The box.)* A search through the boxes stops (Theorem 23.10). Open: a bound on the box accepted, from the list and the direction, which the proof does not give; whether one unit of each active letter and $T \leq 3$ always suffice, or the first ray that needs more; the wall degree of the certificate against the strength from which it integrates (Remark 23.7). [not yet computed]

- **O22** *(The one-step system read.)* The gauge end (Theorem 21.1) and positivity (Theorem 18.5) on $V(k-1) = B(k)V(k)$ directly, without the scalar law: the density vector of the masters and its germ at $f^\star$. It would spare the scalar law, whose degree is the projection's (377 for V at its lean, against entries of degree at most 83). [not yet computed]

- **O23** *(The error of the gauge end.)* A proved bound on the truncations of the germ and of the Beta sums in Theorem 21.1, so that the value carries a proved error as the law carries a certificate; the error now given is the difference of two starting strengths (§23.5, What is certain). [not yet computed]

L.5, Receipts. `receipts_s23.zip`, with a `MANIFEST.txt` mapping each numbered statement of this entry to its script; python-flint, numpy, sympy (for the exact three-mark certificate) and the C eliminators `c/elim.c`, `c/elimp.c` (sparse Gaussian elimination modulo a prime, one pass per strength). The program of §23.5 is `rayreader.zip`, with its own `README.md`: `tests/known_rays.py` and `tests/rational_masses.py` give Table 23.14, and `tests/check_run.py` checks a finished run again at primes and points drawn afresh. Closings enter only the checks.

Aliases, last

§23. The construction is integration-by-parts reduction in the manner of Laporta, on the parametric form $\alpha^{\mathbf{m}} K^{-s} \Psi^{-t}$ of Feynman type, with the strength as the dimension-like parameter: the masters are master integrals, the one-step matrix is the dimension-shift (Tarasov) recurrence written as a first-order difference system, and the law is its reduction to a scalar recurrence by a cyclic vector. The dimension of the ray is the Euler characteristic count of master integrals (Bitoun, Bogner, Klausen, Panzer; Lee and Pomeransky), here Thm. 35.19 of the handbook. The certificate is a creative-telescoping certificate, and the reduction of a form modulo exact forms with poles along the letters is

Griffiths–Dwork reduction in twisted (Aomoto) cohomology. The extra solutions of Proposition 23.4 are the evaluations at singular points of the zero set of the Laurent polynomial, which make the space of its constant-term relations larger than its twisted cohomology. The wall degree is the tropical degree of the insertion against the tropicalization of the stride. The spectrum at infinite strength is the characteristic polynomial of the leading term of a difference system (Birkhoff's characteristic roots), and Proposition 23.11 is the statement that the critical points of $\log f$ diagonalize it, the Morse critical points as eigenvectors. A change of masters is a gauge transformation of the difference system; making its leading term finite is a reduction in the manner of Moser (for difference systems, Barkatou), and the condition $T(k-1)T(k)^{-1} \to 1$ is what keeps the characteristic roots readable from the eigenvalues. The number of critical points of a generic master function is the Euler characteristic of the complement of the letters (Varchenko and Orlik–Terao for arrangements, Huh for smooth very affine varieties), and the exceptional directions of Thm. 35.19(i) are exponents at which critical points are lost; the continuation in the exponents is a parameter homotopy. Modular elimination with rational reconstruction is the method of finite-field reconstruction (as in FiniteFlow and FireFly), and reading a vector of rationals through one common denominator is vector rational reconstruction. The search of Theorem 23.10 is Laporta's algorithm with its range of seeds grown until the targets reduce, and its stopping is the finite dimension of the twisted cohomology; dropping the relations that the targets do not need is the selection of equations in Laporta solvers (as in Kira). The gauge end is the inverse Mellin transform of the recurrence, a Fuchsian equation for the density, whose solution at the dominant singularity is fixed by the Laplace (saddle-point) asymptotics at the native gauge. The masters' values at the gauges spanning is the nondegeneracy of a Vandermonde-type matrix of critical points.